Assignment 2: Relationships, pictures and probability

Statistical Thinking for Data Science • First Semester, 2026/2027

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Covering Weeks 4 to 7 • describing variables, relationships, honest figures, probability models

How to hand this in

Work on your own. Show your working: an answer with no working earns no marks, and a correct number explained badly earns less than a wrong number reasoned carefully. Write your **name, index number and programme** at the top of the first page.
Submit a single PDF through the link announced in class, and bring the paper copy to the next session. **The deadline will be communicated in class and posted on the course page.** Marked out of 50.
If you used any AI tool, add one line at the end saying which and what for. That line costs you nothing. An undeclared tool is academic dishonesty.

Part A: two variables (20 marks)

1. Of 140 students, 70 attended the revision sessions and 70 did not. Among those who attended, 63 passed; among those who did not, 44 passed.

   (a) Build the two-way table and compute the pass rate in each group. (3)

   (b) Compute the risk difference, the risk ratio and the odds ratio, and say in words what each one means here. (5)

   (c) A colleague concludes that the revision sessions raise the pass rate by 27 percentage points. Give two reasons to doubt the causal claim, and describe the study that would settle it. (4)

2. Eight students report weekly laboratory hours x and a practical score y:

x	2	3	4	5	6	7	8	9
y	14	17	16	21	24	22	28	30

   (a) Compute S_{xx}, S_{yy} and S_{xy}, then the correlation. (5)

   (b) Interpret the correlation in one sentence that a first year student would understand. (1)

   (c) A ninth student reports 20 hours and scores 12. Recompute the correlation and explain what has happened. (2)

Part B: pictures (10 marks)

3. For each of the following, name the specific problem, say which way it misleads the reader, and describe the chart you would draw instead. (6)

   (a) A bar chart of mean output for three shifts, with no indication of spread, from eight days each.

   (b) A chart of availability with the vertical axis running from 94% to 100%, shown to the board.

   (c) A bubble chart in which a quantity that doubled is drawn with twice the radius.

4. Draw, by hand, the figure you would use to compare the distribution of practical scores across three programmes with about thirty students each. Label the axes and say in one line what comparison the figure makes. (4)

Part C: probability (20 marks)

5. For each situation, name the distribution you would use and justify it by the *mechanism*, not by the shape of any histogram. (3)

(a) The number of defective sensors in a crate of 15, each independently defective with probability 0.12.

(b) The number of trucks arriving at a weighbridge in ten minutes, at a steady average rate.

(c) The total mass of a loaded truck, made up of many small contributions.

6. Fifteen sensors are tested, each defective with probability 0.12 and independently of the others. Find the probability that none is defective and that three or more are, and state the mean and standard deviation of the number defective. (6)

7. Flaws occur in a weld at an average of 3.2 per metre. Find the probability of no flaw in a metre and of six or more in a metre. State the assumption that lets you treat this as Poisson, and one realistic way it could fail. (6)

8. Compressive strengths are modelled as normal with mean 48 MPa and standard deviation 6 MPa. Find the proportion above 60 MPa, the proportion between 40 and 55, and the strength exceeded by only 5% of cubes. Then give one reason the normal model might be wrong in the lower tail. (5)

Total: 50 marks. Show every intermediate quantity.